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Chapter 1: Introduction and Scope

1.1 Project Focus

This section of JalSanket covers the IoT hardware and embedded system. It is an ESP32-based water-quality monitoring setup using temperature, TDS and turbidity sensors, with an LCD, status LEDs and a buzzer proposed as local indicators.

1.2 Scope of this Report

This report covers only the IoT hardware and embedded-system section. The chlorination and dosing-pump feature has been excluded from the present project scope to keep implementation practical. Advanced water purification can be mentioned separately under future scope.

1.3 Implementation Status

Table 1.1 shows what is present in the supplied firmware and what is only proposed.

Table 1.1: Implementation Status

Item	Status	Description
ESP32 reads DS18B20 on GPIO 4	IMPLEMENTED	1-Wire digital temperature measurement; firmware configured for 10-bit resolution.
TDS module on GPIO 34	IMPLEMENTED	Analog reading with temperature compensation in firmware.
Turbidity module on GPIO 35	IMPLEMENTED	Analog reading and experimental NTU estimate using programmed calibration endpoints.
Wi-Fi and NTP time sync	IMPLEMENTED	Connects to Wi-Fi and obtains time using NTP with IST offset.
Firebase RTDB upload	IMPLEMENTED	Live data to SENSOR_DATA (PUT) and history to SENSOR_HISTORY (POST), about every 3 seconds.
16x2 I2C LCD	PROPOSED	Not present in the supplied firmware; proposed for displaying readings locally.
Status indicator LEDs	PROPOSED	Suggested for power/system, Wi-Fi connection and water-quality alert indication.
Buzzer	PROPOSED	Suggested for an audible alert when a configured warning condition occurs.
pH / free-chlorine / flow / level sensors	NOT INCLUDED	Not part of the current hardware implementation.

Status key: IMPLEMENTED = present in supplied firmware; PROPOSED = planned addition; REQUIRES VERIFICATION = must be checked on the physical hardware.

Chapter 2: Hardware Architecture

The three sensors measure water parameters and send their signals to the ESP32. The ESP32 processes the readings, displays them locally on the proposed LCD, and uploads them to Firebase through Wi-Fi. LEDs and a buzzer can provide simple local status and warning indications. Fig. 1 shows the architecture.

Data flow: DS18B20 + TDS + Turbidity → ESP32 → Processing → LCD / LEDs / Buzzer → Wi-Fi → Firebase RTDB → Website Dashboard.

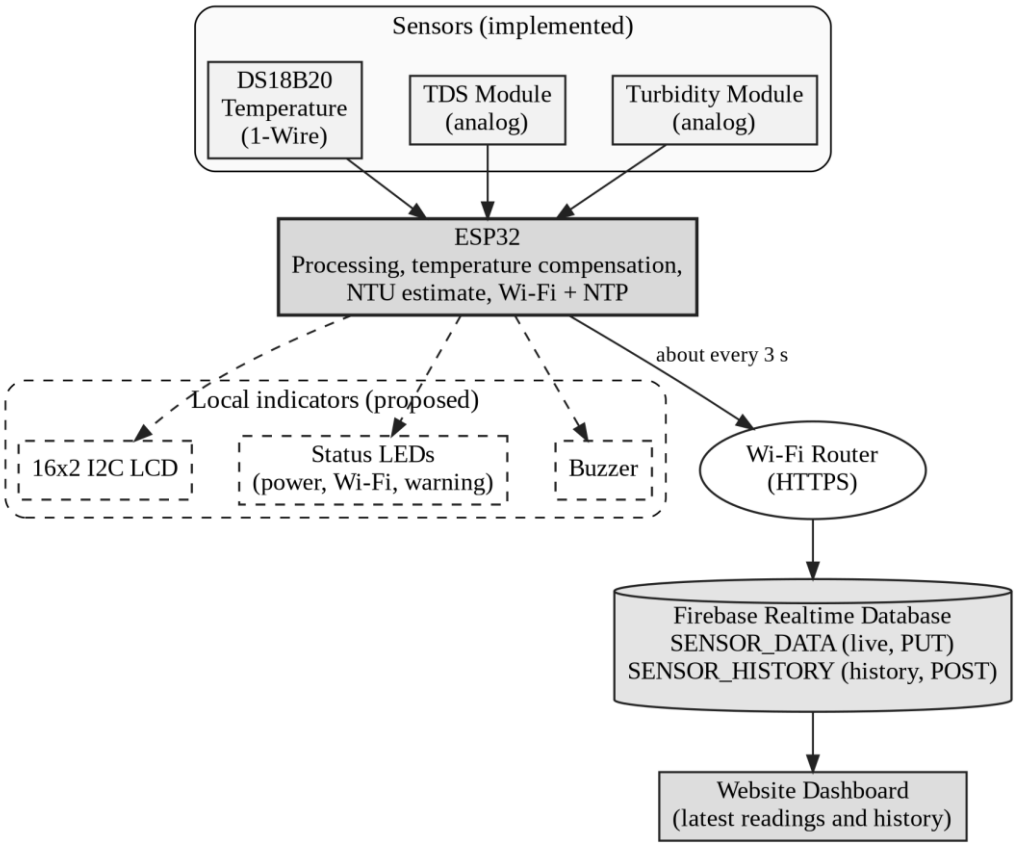


Fig. 1: Hardware Block Diagram and Data Flow

Chapter 3: Components and Key Specifications

Component	Purpose / interface	Supply / key notes
ESP32 development board	Main controller; 1-Wire, ADC, I2C, Wi-Fi and GPIO.	Board supply via USB/5V input; GPIO logic is 3.3V. Exact board variant to confirm.
DS18B20 waterproof probe	Measures water temperature; digital 1-Wire on GPIO 4.	Typically 3.0–5.5V; configured 10-bit resolution gives 0.25°C steps. Accuracy is model/datasheet dependent.
TDS sensor module	Estimates dissolved-solids level from an analog signal to GPIO 34.	Supply and range depend on exact kit. Calibrate with standard solutions; do not assume accuracy without model details.
Turbidity sensor module	Optical sensing; analog output to GPIO 35.	Supply and range depend on exact module. Current firmware NTU is an experimental estimate, not a certified measurement.
16x2 I2C LCD (proposed)	Displays temperature, TDS and turbidity; SDA GPIO 21, SCL GPIO 22.	Backpack commonly uses 5V, but verify module. Ensure I2C lines are not pulled up to 5V at ESP32.
3 indicator LEDs (proposed)	Power/system, Wi-Fi and warning indication.	One current-limiting resistor per LED; proposed GPIOs are listed in Chapter 4.
Active buzzer (proposed)	Audible warning for threshold or error condition.	Use a transistor driver if buzzer current exceeds safe GPIO capability; verify buzzer voltage and current.
4.7 kΩ resistor	DS18B20 DATA pull-up to 3.3V.	Required for a typical 1-Wire setup; verify it is physically installed.
Power supply / wiring	Powers ESP32 and sensor modules.	Use a stable, regulated supply; share signal ground. Keep motor/pump loads out of this design.

Note: The exact TDS and turbidity module models have not been identified in the supplied information. Their operating voltage, measuring range and accuracy must be copied from the actual module datasheets or verified by calibration; they are intentionally not guessed.

Chapter 4: Pin Connections

Table 4.1 lists the pin connections and Fig. 2 shows them around the ESP32. Solid lines are implemented connections and dashed lines are proposed additions.

Table 4.1: Pin Connection Table

Component / signal	ESP32 pin	Status
DS18B20 DATA	GPIO 4	Implemented
TDS analog output	GPIO 34 (ADC1)	Implemented
Turbidity analog output	GPIO 35 (ADC1)	Implemented
LCD SDA	GPIO 21	Proposed
LCD SCL	GPIO 22	Proposed
Power/status LED	GPIO 18 via resistor	Proposed
Wi-Fi status LED	GPIO 19 via resistor	Proposed
Warning LED	GPIO 23 via resistor	Proposed
Buzzer control	GPIO 26 via suitable driver if needed	Proposed

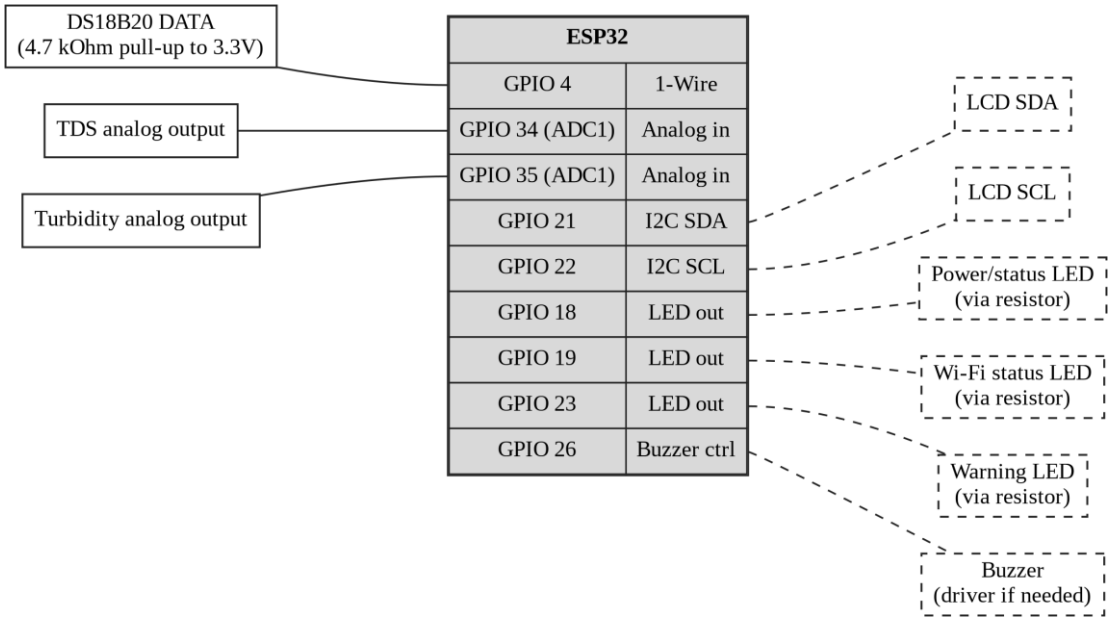


Fig. 2: ESP32 Pin Connection Diagram

Chapter 5: Working of the Circuit

1. **Power ON:** the regulated supply powers the ESP32 and sensor modules, and the ESP32 starts the firmware.
2. **Temperature measurement:** the DS18B20 measures water temperature and sends a digital value to GPIO 4 using 1-Wire communication.
3. **TDS measurement:** the TDS module provides an analog voltage. The ESP32 reads it through GPIO 34 and calculates an estimated TDS value, with temperature compensation applied in the supplied firmware.
4. **Turbidity measurement:** the turbidity module provides an analog voltage to GPIO 35. Firmware converts it into an experimental NTU estimate using its programmed calibration endpoints.
5. **Local indication (proposed):** the 16x2 LCD can show temperature, TDS and turbidity. LEDs can indicate power, Wi-Fi status and warning conditions, and a buzzer can sound for a warning or a sensor/network fault.
6. **Cloud upload:** the ESP32 connects to Wi-Fi, synchronises time using NTP, and sends live and historical readings to Firebase Realtime Database over HTTPS.
7. **Website display:** the website reads the Firebase data and displays the latest readings and history on its dashboard.

The overall working sequence is shown in Fig. 3.

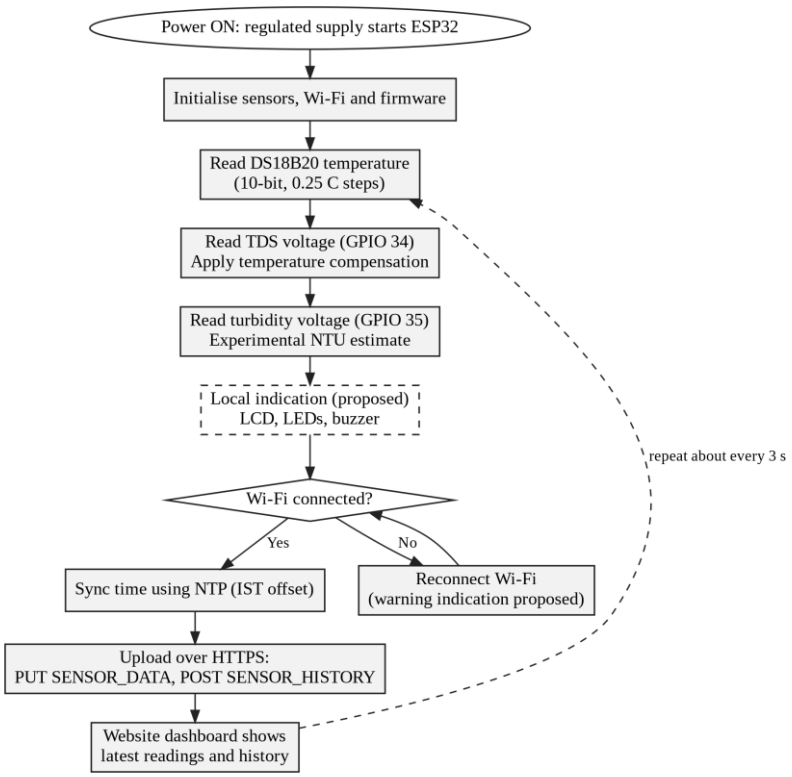


Fig. 3: Working Flowchart of the Circuit

Chapter 6: Communication Used

Interface	Used for
1-Wire	DS18B20 temperature sensor to ESP32 GPIO 4.
Analog ADC	TDS module to GPIO 34; turbidity module to GPIO 35.
I2C (proposed)	LCD SDA to GPIO 21 and SCL to GPIO 22.
GPIO (proposed)	ESP32 controls indicator LEDs and the buzzer driver.
Wi-Fi + HTTPS	ESP32 communicates with Firebase through the wireless router.
Firebase RTDB	Stores live sensor readings and historical records for the website dashboard.

Chapter 7: Circuit Precautions

- ESP32 GPIO pins are 3.3V logic. Do not allow sensor analog outputs or I2C pull-ups to exceed 3.3V.
- Fit a 4.7 kΩ pull-up between DS18B20 DATA and 3.3V.
- Use a separate current-limiting resistor for each LED.
- Drive a buzzer through a transistor if its current requirement is higher than the GPIO can safely supply.
- Use a regulated supply and connect grounds appropriately.
- Verify the LCD I2C address and voltage before connecting it.

Chapter 8: Limitations and Future Scope

8.1 Limitations

The present system monitors temperature, estimated TDS and estimated turbidity. Sensor calibration is necessary for reliable readings. The turbidity-to-NTU conversion is experimental and should be validated using reference standards. TDS and turbidity alone cannot confirm that water is safe to drink.

8.2 Future Scope

pH measurement, free-chlorine residual testing and advanced purification may be considered as future enhancements.

Chapter 9: Conclusion

JalSanket uses an ESP32 to collect water-quality readings from temperature, TDS and turbidity sensors, process the data and upload it to Firebase through Wi-Fi. A proposed LCD, indicator LEDs and buzzer will make the system easier to monitor locally by displaying readings and providing visual and audible alerts. The current scope focuses on monitoring and alerting; chlorination and dosing hardware are excluded.